

UAV Swing Payload Model

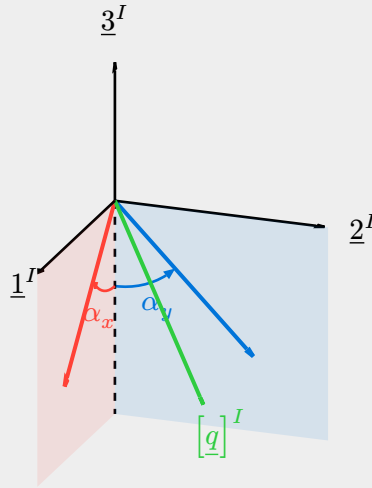
$$m_P \underline{a}_P = T \underline{q} - m_P \underline{g} \quad (1)$$

$$m_D \underline{a}_D = \underline{f} - T \underline{q} - m_D \underline{g} \quad (2)$$

$$\frac{d}{dt} \omega^{BE} = (J_B^B)^{-1} [\underline{n}_B - \underline{\Omega}^{BE} J_B^B \omega^{BE}] \quad (3)$$

$$\underline{s}_P = \underline{s}_D - L \underline{q} \quad (4)$$

Swing angles



$$[\underline{q}]^I = \begin{pmatrix} \cos \alpha_x & 0 & -\sin \alpha_x \\ 0 & 1 & 0 \\ \sin \alpha_x & 0 & \cos \alpha_x \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \alpha_y & -\sin \alpha_y \\ 0 & \sin \alpha_y & \cos \alpha_y \end{pmatrix} [-\underline{3}^I]^I = \begin{pmatrix} \sin \alpha_x \cos \alpha_y \\ \sin \alpha_y \\ -\cos \alpha_x \cos \alpha_y \end{pmatrix} \approx \begin{pmatrix} \alpha_x \\ \alpha_y \\ -1 \end{pmatrix} \quad (5)$$

ArUco Marker Tracking EKF

Frames

$[\cdot]^I$: ENU (+x East, +y North, +z Up)

$[\cdot]^B$: body (+x nose, +y portside, +z up) (6)

$[\cdot]^C$: camera (+x right, +y up, +z optical axis)

State

$$[\xi]^I = \begin{pmatrix} \alpha_x \\ \alpha_y \\ \dot{\alpha}_x \\ \dot{\alpha}_y \\ \psi_P \end{pmatrix} \in \mathbb{R}^5 \quad (7)$$

$$s_x = \sin \alpha_x, \quad c_x = \cos \alpha_x, \quad s_y = \sin \alpha_y, \quad c_y = \cos \alpha_y \quad (8)$$

α_x, α_y : tether swing angles, tipping toward +x^I (East) and +y^I (North) [rad]

$\dot{\alpha}_x, \dot{\alpha}_y$: swing rates [rad/s]

ψ_P : payload yaw: marker-row direction [rad]

Input

$$[T]^{IB} = \begin{pmatrix} c_\theta c_\psi & s_\varphi s_\theta c_\psi - c_\varphi s_\psi & c_\varphi s_\theta c_\psi + s_\varphi s_\psi \\ c_\theta s_\psi & s_\varphi s_\theta s_\psi + c_\varphi c_\psi & c_\varphi s_\theta s_\psi - s_\varphi c_\psi \\ -s_\theta & s_\varphi c_\theta & c_\varphi c_\theta \end{pmatrix} \quad (9)$$

Euler angles: φ : roll θ : pitch ψ : yaw

$$[\underline{a}_D]^I = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = [T]^{IB} [\underline{a}_D]^B \quad (10)$$

$[\underline{a}_D]^I$: acceleration inputs from the drone [m/s²]

Process model

$$\begin{bmatrix} \dot{\xi} \end{bmatrix}^I = f(\xi, a) = \begin{pmatrix} \dot{\alpha}_x \\ \dot{\alpha}_y \\ -\frac{c_x a_1 + s_x a_3}{L c_y} + 2 \frac{s_y}{c_y} \dot{\alpha}_x \dot{\alpha}_y \\ -\frac{c_y a_2 + s_y (c_x a_3 - s_x a_1)}{L} - s_y c_y \dot{\alpha}_x^2 \\ 0 \end{pmatrix} \approx \begin{pmatrix} \dot{\alpha}_x \\ \dot{\alpha}_y \\ -\frac{1}{L} (a_1 + \alpha_x \|g\|) \\ -\frac{1}{L} (a_2 + \alpha_y \|g\|) \\ 0 \end{pmatrix} \quad (11)$$

L : drone tether-pivot to payload CG [m]

$$F = \begin{pmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ -\frac{\|g\|}{L} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\|g\|}{L} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad (12)$$

$F \in \mathbb{R}^{5 \times 5}$: process Jacobian (continuous-time)

Prediction

Jacobian to discrete-time conversion w/ Euler integration.

$$\hat{\xi}_{k+1}^- = \hat{\xi}_k^+ + \Delta t f(\hat{\xi}_k^+, a_s) \quad (13)$$

$$\Phi_k = I_5 + F \Delta t, \quad Q = \text{diag}(0, 0, q, q, q_\psi) \Delta t \quad (14)$$

$$P_{k+1}^- = \Phi_k P_k^+ \Phi_k^\top + Q \quad (15)$$

Δt : time step [s]

$\hat{\xi}_k^-, \hat{\xi}_k^+$: estimate before & after the update at step k

$P_k^-, P_k^+ \in \mathbb{R}^{5 \times 5}$: covariance before & after

Φ_k : discrete time Jacobian

q : process-noise intensity

Shelving Jacobian to discrete-time conversion w/ RK4 for now.

$$k_1 = f(\hat{\xi}_k^+, a_s), \quad k_2 = f\left(\hat{\xi}_k^+ + \frac{\Delta t}{2} k_1, a_s\right) \quad (16)$$

$$k_3 = f\left(\hat{\xi}_k^+ + \frac{\Delta t}{2} k_2, a_s\right), \quad k_4 = f(\hat{\xi}_k^+ + \Delta t k_3, a_s)$$

$$\hat{\xi}_{k+1}^- = \hat{\xi}_k^+ + \frac{\Delta t}{6} (k_1 + 2k_2 + 2k_3 + k_4) \quad (17)$$

$$\Phi_k = I_5 + F \Delta t + \frac{1}{2} (F \Delta t)^2, \quad Q = \text{diag}(0, 0, q, q, q_\psi) \Delta t \quad (18)$$

$$P_{k+1}^- = \Phi_k P_k^+ \Phi_k^\top + Q \quad (19)$$

Δt : time step [s]

$\hat{\xi}_k^-, \hat{\xi}_k^+$: estimate before & after the update at step k

$P_k^-, P_k^+ \in \mathbb{R}^{5 \times 5}$: covariance before & after

Φ_k : discrete time Jacobian

q : process-noise intensity

Geometry

$$[\underline{m}]^I = \begin{pmatrix} \cos \psi_P \\ \sin \psi_P \\ 0 \end{pmatrix}, \quad \text{assuming no roll or pitch} \quad (20)$$

$[\underline{m}]^I$: unit vector along the marker row

Measurement model, $j \in V_k$

$$[p_j]^I = L_m [q]^I - o_j [\underline{m}]^I \quad (21)$$

V_k : marker IDs detected in frame k

L_m : drone tether-pivot to marker-board CG

o_j : marker offset from center along the row, +left

$[p_j]^I$: position of marker j

$$[p_j]^C = \begin{pmatrix} X_j \\ Y_j \\ Z_j \end{pmatrix} = [C]^{CB} \left([\ell]^B + [C]^{BI} [p_j]^I - [t_{BC}]^B \right) \quad (22)$$

$[t_{BC}]^B$: camera optical center (offset due to science stick) [m]

$[\ell]^B$: tether pivot point offset [m]

$$h_j(\xi) = \begin{pmatrix} f_u X_j / Z_j + u_0 \\ f_v Y_j / Z_j + v_0 \end{pmatrix} \quad (23)$$

f_u, f_v, u_0, v_0 : intrinsics [px]

h_j : predicted pixel in image plane

$$H_j = \frac{\partial h_j(\xi)}{\partial [p_j]^C} \frac{\partial [p_j]^C}{\partial [p_j]^I} \frac{\partial [p_j]^I}{\partial [\xi]^I} \quad (24)$$

$$H_j = \begin{pmatrix} \frac{f_u}{Z_j} & 0 & -\frac{f_u X_j}{Z_j^2} \\ 0 & \frac{f_v}{Z_j} & -\frac{f_v Y_j}{Z_j^2} \end{pmatrix} [C]^{CB} [C]^{BI} \left(L_m \frac{\partial [\hat{n}]^I}{\partial \xi} - o_j \frac{\partial [\hat{m}]^I}{\partial \xi} \right) \quad (25)$$

$H_j \in \mathbb{R}^{2 \times 5}$: Jacobian of h_j w.r.t. ξ , by chain rule

$[\ell]^B, [t_{BC}]^B$ are constant

Update

$$z^{(j)} = h_j(\xi) + \varepsilon^{(j)} \quad (26)$$

$m_k \in \{0, 1, 2, 3\}$: number of markers spotted in a given frame

$z^{(j)}$: pixel logged for marker j [px]

$\varepsilon^{(j)}$: measurement noise

$$z_k = \begin{pmatrix} z^{(1)} \\ \vdots \\ z^{(m_k)} \end{pmatrix}, \quad h_k = \begin{pmatrix} h_1 \\ \vdots \\ h_{m_k} \end{pmatrix}, \quad H_k = \begin{pmatrix} H_1 \\ \vdots \\ H_{m_k} \end{pmatrix}, \quad R_k = (\sigma_{\text{det}}^2 + f_u^2 \sigma_\theta^2) I_{2m_k} \quad (27)$$

$$z_k = h_k(\xi) + \varepsilon_k, \quad \varepsilon_k \sim \mathcal{N}(0, R_k) \quad (28)$$

$$y_k = z_k - h_k, \quad S_k = H_k P_k^- H_k^\top + R_k, \quad K_k = P_k^- H_k^\top S_k^{-1} \quad (29)$$

$$\hat{\xi}_k^+ = \hat{\xi}_k^- + K_k y_k \quad (30)$$

$$P_k^+ = (I_5 - K_k H_k) P_k^- \quad (31)$$

y_k : innovation S_k : innovation covariance $K_k \in \mathbb{R}^{5 \times 2m_k}$: Kalman gain
 σ_{det} : marker detection noise [px] σ_θ : attitude 1σ [rad]

If no markers detected:

$$m_k = 0 \implies \hat{\xi}_k^+ = \hat{\xi}_k^-, \quad P_k^+ = P_k^- \quad (32)$$